

ECON 609: Heterogeneous Agent Models

Minsu Chang

Georgetown University

Lecture # 11

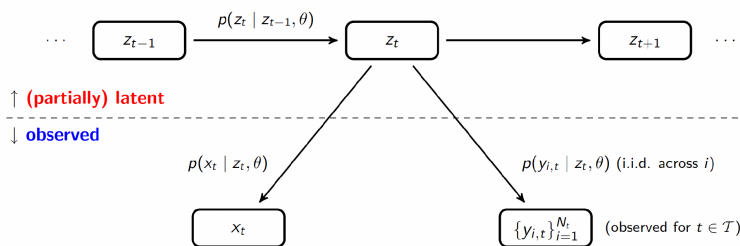
Introduction

- ▶ Attempt to use micro data to estimate HA model: Liu and Plagborg-Møller (QE, 2022).
- ▶ Transition dynamics with sequence space Jacobian approach: Auclert et al. (ECMA, 2021)

Liu and Plagborg-Møller (QE, 2022)

- ▶ This paper provides a full-information inference for heterogeneous agent models.
- ▶ Challenge: latent aggregate states affect cross-sectional distributions.
- ▶ They develop a generic method
 - ▶ Unbiased likelihood estimate $\rightarrow$ fully efficient and valid Bayes inference.
 - ▶ Optimally exploit macro and micro data.

Setting



- z_t : (partially) latent aggregate states (includes param of cross-sec dist.)
- x_t : observed macro time series
- $y_{i,t}$: observed micro data, repeated cross sections (i.i.d. across i and independently across t , given aggregate states)

Example: Heterogeneous Household Model

- ▶ Households $i \in [0, 1]$:

$$\max_{c_{i,t}, n_{i,t}} E_0 \left[\sum_{t=0}^{\infty} \beta^t (\log c_{i,t} - \chi n_{i,t}) \right]$$

$$\text{s.t. } c_{i,t} = W_t \lambda_i n_{i,t}.$$

- ▶ Permanent heterogeneity:

$$\log(\lambda_i) \sim N(\mu_\lambda, \sqrt{-2\mu_\lambda}) \Rightarrow E[\lambda_i] = 1.$$

- ▶ Representative firm: $X_t = Z_t N_t$,

$$\log Z_t = \rho \log Z_{t-1} + \epsilon_t, \quad \epsilon_t \sim_{i.i.d.} N(0, 1).$$

- ▶ **Latent** aggregate state: $z_t = \log Z_t$.

Heterogeneous Household Model

- **Observed** macro variable: $x_t = \log X_t + \eta_t$, $\eta_t \sim_{i.i.d.} N(0, \sigma_e^2)$ (measurement error).

$$(x_t | z_t, \theta) \sim (z_t - \log \chi, \sigma_e^2).$$

- **Observed** micro cross-sections: $y_{i,t} = \log c_{i,t}$.

$$(y_{i,t} | z_t, \theta) \sim (z_t - \log \chi + \mu_\lambda, \sqrt{-2\mu_\lambda}).$$

- **Parameters**: $\theta = (\beta, \chi, \mu_\lambda, \rho, \sigma_e)$.

Unbiased Likelihood Estimate

$$\begin{aligned} p(x, y|\theta) &= \underbrace{p(x|\theta)}_{\text{macro}} \underbrace{p(y|x, \theta)}_{\text{micro}} = p(x|\theta) \int p(y|z, \theta) p(z|x, \theta) dz \\ &= p(x|\theta) \int \prod_{t \in \mathcal{T}} \prod_{i=1}^{N_t} p(y_{i,t}|z, \theta) p(z|x, \theta) dz. \end{aligned}$$

- ▶ Integral in micro likelihood is usually hard to compute.
- ▶ They compute instead a numerically unbiased estimate:

$$\int \prod_{t \in \mathcal{T}} \prod_{i=1}^{N_t} p(y_{i,t}|z_t, \theta) p(z|x, \theta) dz \approx \frac{1}{J} \sum_{j=1}^J \prod_{t \in \mathcal{T}} \prod_{i=1}^{N_t} p(y_{i,t}|z_t = z_t^{(j)}, \theta).$$

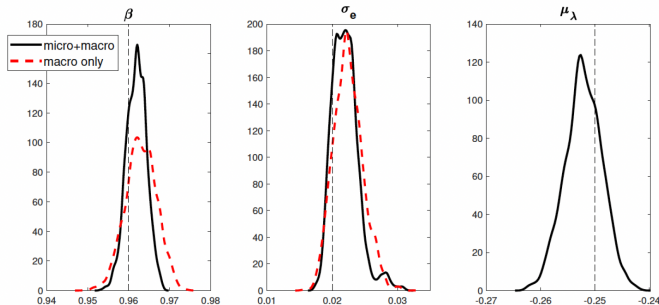
- ▶ $\{z_t^{(j)}\}$ are drawn from $p(z|x, \theta)$ based on Kalman filter.

Shocks

- ▶ They use a model similar to Krusell-Smith (1998).
- ▶ Aggregate productivity shock
- ▶ Household:
 - ▶ Idiosyncratic employment risk and asset.
 - ▶ Permanent individual productivity: λ_i is log-normally distributed where $E[\log \lambda_i] = \mu_\lambda$ and $E[\lambda_i] = 1$.
- ▶ Not in our class, but also heterogeneous firm model á la Khan and Thomas (2008).

Estimated Parameters

- β : household discount factor
- σ_e : stdev of the measurement error in log aggregate output
- μ_λ : parameter of individual productivity distribution



- ▶ Micro data can sharpen inference.
- ▶ Micro data can be essential for identification μ_λ .

Auclert et al. (ECMA, 2021)

- ▶ Today, we will consider transitions in Aiyagari economies.
- ▶ Transitions can be hard to compute, but not always.
- ▶ Today: MIT shocks in Aiyagari economies.
 - ▶ An “MIT shock” is an unexpected shock that hits an economy at its steady state, leading to a transition path back towards the economy’s steady state.

Transitions in Aiyagari

- ▶ Households: assets a and productivity x

$$V_t(a, x) = \max_{c, h, a'} \left\{ \frac{c^{1-\sigma}}{1-\sigma} - \frac{h^{1+\varphi}}{1+\varphi} + \beta \sum_{x'} \pi(x'|x) V_{t+1}(a', x') \right\}$$

$$\text{s.t.} \quad c + a' = w_t x h + (1 + r_t) a, \quad a' \geq \underline{a}.$$

- ▶ Firms: $w_t = F_L(K_t, L_t)$, $r_t + \delta = F(K_t, L_t)$, and $F(K_t, L_t) = K_t^{1-\alpha} L_t^\alpha$.
- ▶ Equilibrium: $L_t = \int x h_t(a, x) d\Phi_t(a, x)$ and $K_t = \int a'_t(a, x) d\Phi_t(a, x)$.

Transitions in Aiyagari: iterate on sequences

1. Guess a sequence of prices $\{w_t, r_t\}_t$ for $t = 1, \dots, T$.
2. Given a final $V_T(a, x)$, solve backward: $\{V_t(a, x), \mathbf{a}'_t(a, x)\}_t$
3. Solve forward: $\Phi_{t+1} = Q_t \Phi_t$ with Q_t transition function.

$$L_t^s = \sum_{a,x} x \mathbf{h}_t(a, x) \Phi_t(a, x), \quad K_t^s = \sum_{a,x} \mathbf{a}'_{t-1}(a, x) \Phi_t(a, x).$$

4. Compute errors/implied prices with firms' FOC

$$\hat{w}_t = F_L(K_t^s, L_t^s), \quad \hat{r}_t + \delta = F_K(K_t^s, L_t^s).$$

5. Update $\{w_t, r_t\}_t$ and iterate.

Transitions in Aiyagari: iterate on sequences

- ▶ Iteration on sequences works well in simple/stable models.
- ▶ Not always easy to get implied “ $\hat{p}$ ” price
- ▶ Useful alternative: Quasi-Newton algorithm

Transitions in Aiyagari: Quasi-Newton

- ▶ For given prices $\vec{p} = \{w_t, r_t\}_t$, compute errors as excess demand
 - ▶ Firm's demand: $L_t^d = L(w_t, r_t)$ and $K_t^d = K(w_t, r_t)$
 - ▶ Excess demand for labor and capital: $E_t^L = L_t^d - L_t^s$ and $E_t^K = K_t^d - K_t^s$.
- ▶ Vector with all errors:
$$E(\vec{p}) = [E_1^L, E_2^L, \dots, E_T^L, E_1^K, E_2^K, \dots, E_T^K]'$$
- ▶ Solution: $E(\vec{p}) = 0$.

Transitions in Aiyagari: Quasi-Newton

- ▶ Quasi-Newton step:

- ▶ $E(\vec{p}) \approx E(\vec{p}_0) + J(\vec{p}_0)(\vec{p} - \vec{p}_0)$ with $J(\vec{p}_0) = \nabla E(\vec{p}_0)$.

- ▶ Iteration: $E(\vec{p}) = 0 \rightarrow \vec{p}_{n+1} = \vec{p}_n - J(\vec{p}_n)^{-1} E(\vec{p}_n)$

- ▶ Jacobian $J(\vec{p}_n)$ at $\vec{w} = \vec{w}_{ss}$ and $\vec{r} = \vec{r}_{ss}$.

- ▶ Set $\vec{w}_\tau = [w_{ss}, w_{ss}, \dots, \underbrace{w_{ss}(1 + \epsilon)}_{\tau\text{-th entry}}, \dots, w_{ss}]$. Similar for $\vec{r}_\tau$.

- ▶ Compute Jacobian in a “brute force” way

$$J(:, \tau) = \frac{E(\{\vec{w}_\tau, \vec{r}_{ss}\}) - E(\{\vec{w}_{ss}, \vec{r}_{ss}\})}{w_{ss}\epsilon}$$
$$J(:, T + \tau) = \frac{E(\{\vec{w}_{ss}, \vec{r}_\tau\}) - E(\{\vec{w}_{ss}, \vec{r}_{ss}\})}{r_{ss}\epsilon}$$

Transitions in Aiyagari: Faster J computation

- Revisit Bellman equation

$$V_{\mathbf{t},\tau}(a, x) = \max_{c, h, a'} \left\{ \frac{c^{1-\sigma}}{1-\sigma} - \frac{h^{1+\varphi}}{1+\varphi} + \beta \sum_{x'} \pi(x'|x) V_{t+1}(a', x') \right\}$$

$$\text{s.t.} \quad c + a' = w_t x h + (1 + r_t) a, \quad a' \geq \underline{a}.$$

- Say $r_t = r_{ss}$ and $\vec{w}_\tau$ (w changes in period τ only).
- Want to compute $V_{t,\tau}$ and $\mathbf{a}'_{t,\tau} \rightarrow$ enough to compute J .
 - For $t \geq \tau + 1$: $V_{t,\tau} = V_{ss}$ and $\mathbf{a}'_{t,\tau} = \mathbf{a}'_{ss}$.
 - Focus on $V_{\tau-j,\tau}$ and $\mathbf{a}'_{\tau-j,\tau}$ for $j = 0, 1, \dots, \tau$. (shock is j period ahead).

Transitions in Aiyagari: Faster J computation

- ▶ Key insight: only distance to τ (shock) matters:

$$\mathbf{a}'_{\tau-j,\tau} = \mathbf{a}'_{\tau-j+\Delta,\tau+\Delta}.$$

- ▶ For J , enough to solve for $\vec{w}_T$ and $\vec{r}_T$ (Auclert et al., 2021)
- ▶ Compute policies for $\vec{w}_T$ (one sequence!) and set

$$a'_{t,\tau} = \begin{cases} a'_{T-(\tau-t),T}, & \text{if } t = 0, \dots, \tau \\ a'_{ss}, & \text{if } t = \tau + 1, \dots, T \end{cases}$$

- ▶ Same for $\vec{r}_T$.
- ▶ Computation time greatly reduced compared to “brute force”.